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Hybrid Optimization-enabled Deep Learning Model for Intrusion Detection in Cloud Computing

Abstract

The multi-tenant nature, dynamic resource provisioning, and wide geographic dispersion of cloud platforms cause major challenges to maintaining reliable security. Conventional Intrusion Detection Systems (IDS) designed for on-premise networks have difficulty adapting to the special characteristics of cloud environments, which limits their ability to identify and prevent attacks. Hence, a scalable, adaptable, and effective IDS technique called Harmonic Secretary Bird Optimization Algorithm_Quasi-Recurrent Neural Network (HSBOA_QRNN) which overcomes the specific difficulties of cloud environments is devised in this article. The first step is to simulate the cloud. Afterward, the input network traffic data is obtained from the given dataset, which is then preprocessed. Data normalization is performed using min-max normalization, and missing value restoration is used for data preprocessing. Following that, Support Vector Machines-Recursive Feature Elimination (SVM-RFE) is used for selecting the most pertinent features, where SVM is optimally trained by employing the proposed HSBOA. Finally, the intrusion is detected using the QRNN, which is also trained by exploiting the HSBOA. The detected outcomes are Denial of Service (DoS), User to Root (U2R), Remote to Local (R2L), Probe, and Normal. Moreover, the HSBOA_QRNN gained supreme accuracy of 91.358%, True Negative Rate (TNR) of 92.353%, and True Positive Rate (TPR) of 91.223%.

Keywords: Cloud Computing, Intrusion Detection, Cyber Security, Deep Learning, Quasi-Recurrent Neural Network.

1. Introduction

"Cloud computing" refers to internet-based computing that provides platforms, software, infrastructure, and resources through virtual shared servers and workstations [1]. Cloud computing services today offer an assortment of features, including large networks and resource pooling [6]. Cloud computing's capacity to provide on-demand, optimized network access and a shared pool of reconfigurable computing resources has allowed it to grow quickly in recent years. Through communication with the service provider or with little management, these resources can be rapidly provisioned and released. These benefits have led to the increasing deployment of cloud computing in more regions and the migration of more applications to cloud environments [2]. These days, the development of digital technologies has resulted in the rapid expansion of cloud computing applications across various industries because of its services and benefits, including low cost, availability, and expandability [10] [8]. Cloud computing is utilized in public, private, and hybrid cloud deployment frameworks. There are several models of cloud computing, including Platform as a Service (PaaS), Software as a Service (SaaS), and Infrastructure as a Service (IaaS) [7]. The privacy and safety of the cloud resources and services being deployed are also gaining more attention as cloud services are delivered internet [11] [2]. Cloud technology is expanding quickly today with an exponential rise in the number of cloud service providers, users, and threats targeting providers, and users. As a result, there are numerous security concerns pertaining to integrity, confidentiality, availability, and accessibility, when users upload sensitive data to cloud storage. Furthermore, the continuous provision of cloud services enhances the possibility of intrusion in the absence of perfect protection [6].

The Internet is a foremost source of risks that can infect cloud resources and systems because it makes it easier to access the services provided by the cloud. The open and widely distributed nature of the cloud structure makes it an easy target for intruders [14] [6]. Cloud computing's cyber security concerns encompass issues with individual parts of its complicated architecture. Cloud computing encompasses unique information processing technologies, whereby the computing system's resources are presented as an Internet service based on the requirements of the user [9]. One of the most significant and continuing cyber security problems is the identification of hostile network activity [1]. Many techniques have been devised and implemented to protect cloud environments, data, and applications from threats, like firewalls and antivirus software, but they still require improvement [12] [3]. The complication and scale of the distributed infrastructure of cloud computing networks necessitate the use of advanced and modern technologies in order to evaluate the huge amount of data formed from the interactions between those devices. The cloud platform offers Cyber Security Systems (CSS) devices and methods to detect cloud setup attacks. An IDS is a commonly used component of computer system security procedure that protects cloud environments from online threats [13] [1]. The main causes of intrusion include attackers attempting to access cloud resources online, authorized users attempting to acquire privileges not officially granted to them, and users abusing their access rights [6].

A computer or network's activity is monitored by Intrusion Detection, which observes for indications of an intrusion. IDS may be software, hardware, or a combination of both that automates the Intrusion Detection process [6]. Attacks from outside the scheme are detected by IDS. IDS is essential for identifying a variety of attack methods. The primary focus of IDS can be considered a classification problem. There are numerous attack types that can be used against IDS, including normal, R2L, U2R, probe, and DOS [4]. IDS gathers information from the network or from being observed, and it notifies the network manager by emailing or recording intrusion events. IDS performance is impacted by false positives and negatives, so the warnings they issue are rarely connected to actual intrusion [6]. Hybrid detection, anomaly detection, and misuse detection are the three categories of IDS [3]. The anomaly detection technique finds intrusions using unknown attacks, while the misuse detection method finds intrusions employing known attacks. The benefits of these two approaches are combined to create the hybrid approach [15]. Even though there are more solutions available for secure cloud environments, the latest IDSs have a number of serious drawbacks, such as massive volumes of data analysis, real-time detection, poor data quality, and other issues that aim to reduce detection model performance. These days, academic researchers demonstrate that intelligent learning techniques like ensemble learning [26], DL [27], and Machine Learning (ML) [28] are helpful in a variety of contexts and can carry out network security [16] [7].

The main goal is to create an efficient and adaptable IDS called HSBOA_QRNN that can handle the challenges of cloud environments. First, a cloud simulation is conducted. Thereafter, the network traffic data is retrieved from the given database and preprocessed. This preprocessing involves missing values restoration and normalizing the data exploiting min-max normalization. Then, feature selection is conducted employing SVM-RFE, with the SVM being optimized through the proposed HSBOA. Lastly, intrusion detection is carried out exploiting a QRNN, which is tuned with HSBOA. The HSBOA is introduced by amalgamating SBOA and harmonic analysis. The output of the intrusion detection includes categories such as DoS, Probe, R2L, U2R, and Normal.

This article's primary contribution is listed below.

- **Established HSBOA_QRNN for intrusion detection in cloud computing:** An effectual method called HSBOA_QRNN is devised for detecting intrusion in cloud computing. Here, feature selection is done using SVM-RFE, where the SVM is optimally trained through the proposed HSBOA. Moreover, the intrusions are detected by employing QRNN, which is trained using HSBOA. The HSBOA is established by merging SBOA and harmonic analysis.

The residue of the manuscript is organized beneath: Section 2 analyses the literature on current intrusion detection techniques; Section 3 introduces the IDS system model; Section 4 describes the HSBOA_QRNN, which was devised for intrusion detection; Section 5 addresses the assessment of the experimental findings; and Section 6 covers the article's conclusion and future work.

2. Motivation

IDS focuses more on proactive threat mitigation than reactive detection as cyber threats become more complex. Conventional IDS were primarily developed to identify attacks after they occurred which frequently resulted in delayed reactions. Using Artificial Intelligence (AI) and machine learning, modern IDS expect, detect, and prevent possible threats before they have a chance to do damage, offering organizations a more proactive security position. Therefore, it inspires the development of HSBOA_QRNN for intrusion detection.

2.1 Literature survey

A technique for Cyber Security in Cloud Computing - Self-Attention Conditional Generative Adversarial Network Optimised with Crayfish Optimization Algorithm (CSCC-SACGAN COA) was developed by Sugitha, G. and Chaluvraj, P.B., [1]. This approach had a lower detection error rate, more scalability, and less computation time. However, the variations in traffic volume and topology made this approach less effective. Long, Z., *et al.* [2] established a Transformer-based network intrusion detection algorithm for cloud security. This approach performed better and was more accurate and precise. Though, this model did not detect complex intrusion patterns in distributed scenarios, like cloud datacentres. Attou, H., *et al.* [3] developed Radial Basis Function Neural Network (RBFNN) and Random Forest (RF) to detect malicious activities in cloud computing. This model used fewer variables, which shortened the prediction time. However, this technique was unable to minimize the complication of the input data. Samunnisa, K., *et al.* [4] introduced an Efficient Hybrid clustering and classification model for intrusion detection in cloud computing. This technique was effective in identifying numerous kinds of attacks in the cloud environment. However, this approach was not applied in attack behavior that finest indicates the specifics of each attack.

Kavitha, C., *et al.* [5] presented Filter-based Ensemble Feature Selection (FEFS) and Deep Learning Model (DLM) for intrusion detection in cloud computing. The technique demonstrated good performance despite the small amount of R2L and U2R cases available. Nevertheless, this model was unable to enhance the accuracy of detection. Alohal, M.A., *et al.* [6] established Improved Metaheuristics with a Fuzzy Logic-based Intrusion Detection System for the Cloud Security (IMFL-IDSCS). This approach produced the highest level of security and detection

74 efficiency. However, this approach was not effective in detecting ransomware, malware, or other types of attacks in the Android environment. Attou, H., *et al.* [7] devised Cloud-based intrusion detection model, and Random Forest (RF) for cloud security. This approach had a minimum execution time and used just two features to distinguish among normal and aberrant traffic. However, this approach was unable to utilize ensemble learning and DL approaches for boosting the efficacy of the model. Bakro, M., *et al.* [8] introduced RF for cloud-based intrusion detection. This technique proved effective in identifying intrusions in cloud environments by achieving a lower false alarm rate at a manageable resource cost. Nonetheless, recent databases, which include a wide variety of threats that represent current real networks, were not used to test this approach.

2.2 Major challenges

The different intrusion detection methods in cloud computing face the following difficulties.

- The CybS-CC-SACGAN-COA was established in [1] to improve cloud computing environment security. However, the incorporation of SVM and optimization algorithms for the efficient optimal feature selection was not taken into account by this method.
- The RF-PBFNN [3] model decreased prediction time and increased accuracy rate. However, this model did not involve DL with optimization for enhancing the precision and effectiveness of computer network security threat detection.
- The FEFS-DLM [5] identified and selected the most relevant features that can be used to enhance intrusion detection accuracy. However, this method didn't consider the network traffic and attack behavior for improved performance.
- IMFL-IDSCS [6] handled data management, likely centralizing data access and processing to maintain efficient monitoring and detection. But this technique failed to consider DL with optimization for enhancing the accuracy and efficiency for detecting security threats in computer networks.
- A number of intrusion detection methods have been employed to detect cloud computing intrusions, but their accuracy has not been assured. Furthermore, some approaches have utilized DL frameworks that can generate good outcomes but require important computational resources.

3. System model

66 A network-based IDS (NIDS) that employs sensors to sensitize and monitor network traffic in addition to checking for malicious packets is used in this work, which employs a multi-threaded IDS model [23]. The system alerts a third-party monitoring service to intrusions, and the service can promptly report the incident to the cloud user organization management system and provide the cloud service provider with an advisory report. Cloud computing offers storage and application services on distant servers. Large volumes of data could be processed by the multi-threaded IDS, which could also lower packet loss. A third-party monitoring service would then notify the cloud user directly about the compromised system after the IDS had processed the monitored alerts effectively. In addition, the third-party monitoring service would offer the cloud service provider specialized advice regarding system errors and intrusion vulnerabilities. Through the cloud network, the user can access data stored on remote servers at the service provider's location. A multi-threaded NIDS is used to track and log user requests and actions. 9 With expert guidance for cloud service providers, the alert logs are easily shared with cloud users. Figure 1 portrays the system model for IDS.

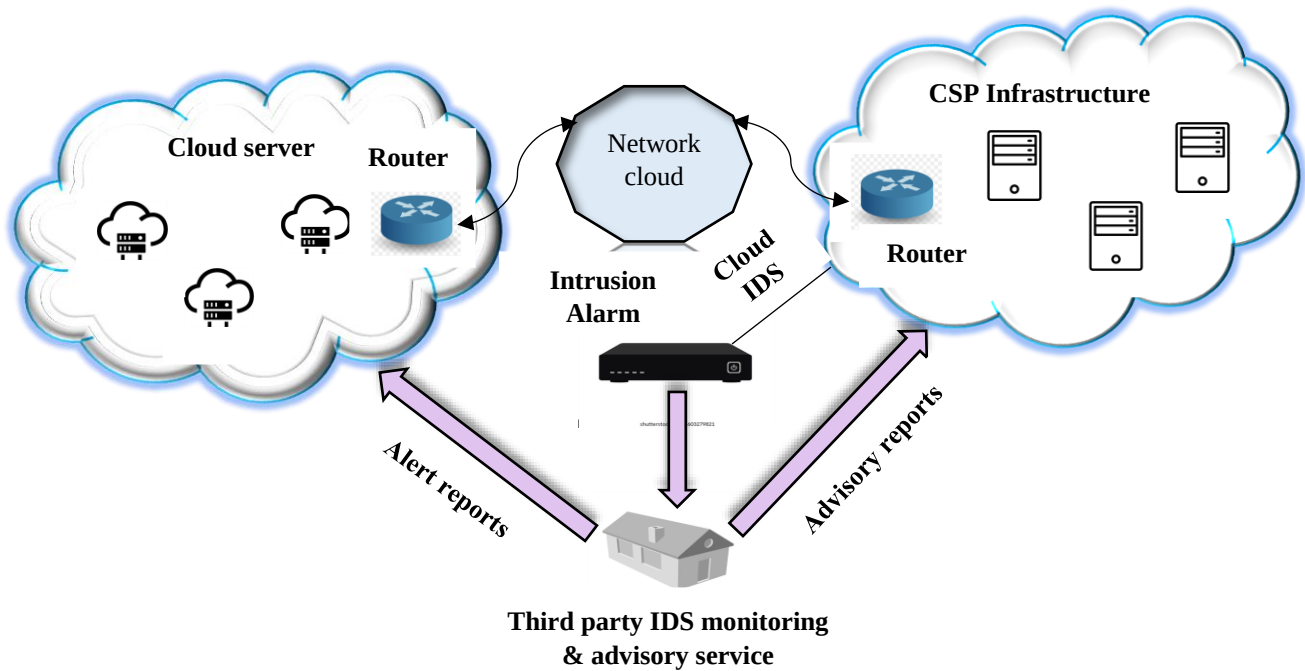


Figure 1. System model for IDS

4. Proposed Harmonic Secretary Bird Optimization Algorithm_Quasi-Recurrent Neural Network for intrusion detection in cloud computing

Cloud computing environments are vulnerable to a variety of security threats due to their growing complexity and scale, which makes effective intrusion detection systems necessary. This article presents an innovative optimization-based DL framework for intrusion detection. Primarily, the cloud simulation is carried out, followed by the acquisition of network traffic data from the given database. This data is then subjected to preprocessing, which involves missing value restoration and data normalization using min-max normalization [17]. Once the preprocessing is completed, feature selection is performed using the SVM-RFE [18] technique, where the SVM is optimally trained through the proposed HSBOA. The HSBOA is newly designed by incorporating the harmonic analysis [22] and SBOA [20]. At last, the intrusion is detected by employing a QRNN [19], which is also tuned with the proposed HSBOA. The detection outcomes are categorized as Probe, DoS, R2L, U2R, or Normal. The graphical representation of the HSBOA_QRNN for intrusion detection in cloud computing is signified in Figure 2.

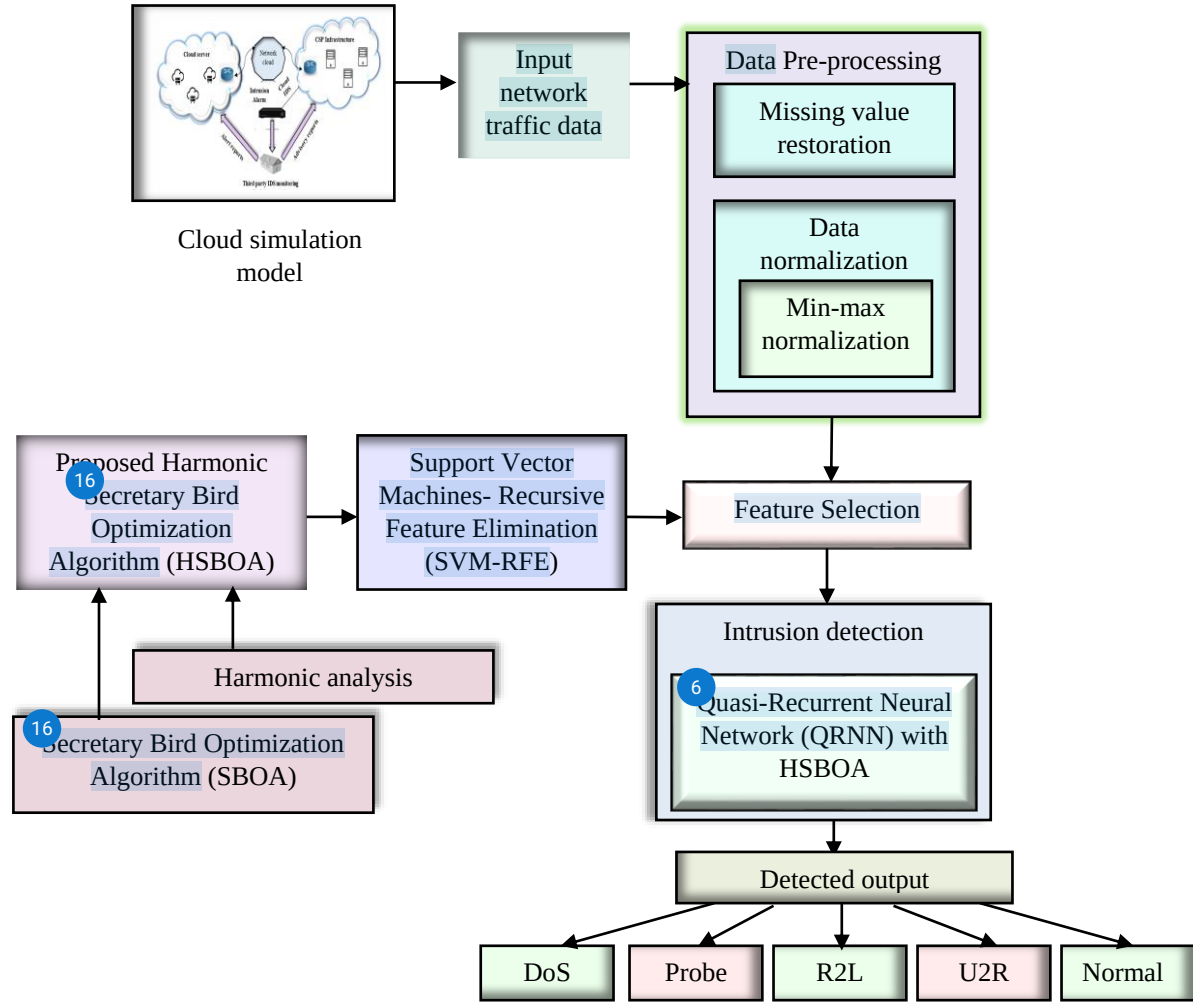


Figure 2. Systematic view of HSBOA_QRNN for intrusion detection in cloud computing

4.1 Data acquisition

The input network traffic data is gathered primarily from the ISCX NSL-KDD dataset 2009 [21]. The collected dataset is specified below,

$$Z = [Z_1, Z_2, \dots, Z_k, \dots, Z_r] \quad (1)$$

wherein, the ISCX NSL-KDD dataset 2009 acquired for the detection intrusion is represented as Z , r specifies overall data records accessible in the database, and Z_k signifies k^{th} record obtained as an input for the detection procedure.

4.2 Data preprocessing

Preparing raw data into a clean and usable format for analysis requires a crucial step called data preprocessing. Preprocessing is essential because it improves model performance, lowers

computational complexity, and minimizes problems with data quality like outliers or missing values. It ensures a fair representation of features and avoids biases caused by imbalanced databases. In this article, the data preprocessing involves two steps, such as data normalization is carried out using min-max normalization [17] and missing value restoration is applied to handle any incomplete data. Moreover, input network traffic data Z_k is taken as an input to data normalization process.

23 4.2.1 Data normalization

The general purpose of data normalization is to transform and standardize unstructured data for arranging it into structured form. The min-max normalization [17] is used in this work to normalize input network traffic data Z_k and convert the data into a specific range. The original data Z_k is transformed linearly into the designated interval (new_{min}, new_{max}) by min-max normalization as follows,

$$Z_k^{norm} = new_{min} + (new_{max} - new_{min}) \times \left(\frac{Z_k - Z_{min}}{Z_{max} - Z_{min}} \right) \quad (2)$$

The data is proportionately scaled from (Z_{min}, Z_{max}) to (new_{min}, new_{max}) using this method. Here, $max(Z_k)$ indicates the attribute's maximum value and $min(Z_k)$ is used to represent the minimum value. Additionally, Z_k^{norm} represents the normalized output obtained by applying Min-Max normalization to input data Z_k . This result is then passed on to the missing value restoration.

4.2.2 Missing value restoration

Here, normalized data Z_k^{norm} is taken as an input for missing value restoration. Missing value restoration is the process of addressing incomplete or missing data in databases by substituting estimated values for missing values. This is an essential component of data preprocessing since many machine learning algorithms depend on complete datasets to perform properly. The outcome attained from missing value restoration is represented as ε_m .

33 4.3 Feature selection

Feature selection is the process of choosing a subset of pertinent attributes from a higher amount of available features in the database. Feature selection reduces noise in the data and produces more accurate predictions by choosing only the most essential features. Redundant features can introduce noise and minimize the technique's capability for generalizing the new data. This paper uses the SVM-RFE [18] technique to sequentially backward eliminate the relevant features from the missing value restoration ε_m . Further, the SVM is optimally trained through the proposed HSBOA, which is newly introduced by combining harmonic analysis and SBOA.

i) SVM-RFE

The SVM-RFE [18] algorithm creates a ranking coefficient to identify the signature attributes based on the weight vector J that SVM creates during training. Following every iteration, the signature feature with lowest ranking coefficient is eradicated. SVM-RFE can efficiently handle complex databases where features have non-linear interactions. By minimizing the amount of

irrelevant or redundant features, SVM-RFE can help enhance the model's accuracy, reduce overfitting, and improve generalization to novel data.

Contemplate training instances $\{p_R, \varpi_R\}$, $\varpi_R \in \{-1, +1\}$ and C denotes the ordering set of output features.

i) Set $\varepsilon_m = 1, 2, \dots, \eta$ as the initial feature set; $c = \emptyset$ is a feature-organized set.

ii) Continue doing these steps until $\varepsilon_m = \emptyset$.

a) Acquire the training set and the candidate feature set

b) Train the SVM classifier to obtain J

c) Determine the ranking norms score:

$$\beta_n = J_n^2, \quad n = 1, 2, \dots, |\varepsilon_m| \quad (3)$$

d) Find the features that have the lowest score on the ranking criteria

$$\arg \min_n \beta_n \quad (4)$$

e) Update feature set $c = \ell \cup c$, where ℓ denotes the lowest-ranked features

f) Remove this feature in ε_m so as $\varepsilon_m = \varepsilon_m / \ell$.

Here, ζ_{fs} stands for the result obtained from SVM-RFE.

3 In two-class classification tasks, the SVM's basic concept is to find the best separating hyperplane in the feature space that optimizes the margin among the training set's positive and negative sample intervals. Specified the group of training samples on a feature space $K = \{(t_1, I_1), (t_2, I_2), \dots, (t_G, I_G)\}$, $I_d \in \{-1, +1\}$, $d = 1, 2, \dots, G$, where the sample size is indicated as G . Therefore, the objective function is expressed as follows,

$$\max \nu \quad (5)$$

Here, the geometric interval is represented as ν . Further, based on the definition of geometric interval ν , the expression can be formulated as follows,

$$I_d (J^T t_d + \lambda) = \nu_d \geq \nu, \quad d = 1, \dots, W \quad (6)$$

where, λ indicates bias and W is the number of samples. Following the derivation of several formulas, the aforementioned objective function is changed into,

$$\max \frac{1}{\|J\|}, \quad I_d (J^T t_d + \lambda) \geq 1, \quad d = 1, \dots, W \quad (7)$$

This problem's solution could yield the SVM classifier. The training sample is unable to meet condition $I_d(J^T t_d + \lambda) \geq 1, d = 1, \dots, W$ when it is linearly inseparable or the separability of the sample is unknown earlier. The generalized classification hyperplane is the optimal hyperplane obtained at this point, and slack variables are added to allow for mis-divided samples. Hence, relaxation term is included, and the expression becomes,

$$I_d(J^T t_d + \lambda) + M_d \geq 1, d = 1, \dots, W \quad (8)$$

Obviously, the error M_d is greater than zero when the division is incorrect. By adding the error penalty coefficient u , the generalized optimum classification surface problem can be expanded to the minimal of the following functions.

$$\min \left(\frac{1}{2} \|J\|^2 + u \left(\sum_{d=1}^G M_d \right) \right), I_d(J^T t_d + \lambda) + M_d \geq 1, d = 1, \dots, W \quad (9)$$

4.3.1 Proposed Harmonic Secretary Bird Optimization Algorithm

The performance of SVM-RFE technique for feature selection is enhanced by applying HSBOA to train the SVM-RFE parameters. In this case, harmonic analysis [22] and SBOA [20] are combined to form HSBOA. A novel population-based metaheuristic algorithm called SBOA [20] imitates the secretary birds' ability to survive in the wild. Secretary birds acquired the ability to hunt and move in order to combat predators. Exploration and exploitation are two phases of survival adaptability in such situations and this serves as SBOA's inspiration. The algorithm's exploration and exploitation phases mimic the actions of secretary birds as they try to protect themselves from predators and catch snakes, explicitly. SBOA accomplishes extremely well in attaining a balance among speed, accuracy, and exploration and exploitation. A function's relationship to its frequency representation is examined through a process called harmonic analysis [22]. For periodic functions, the frequency representation is determined by the harmonic analysis using the Fourier transform. The HSBOA handled complex, multimodal, and high dimensional issues efficiently due to its hybrid nature. Moreover, this technique was simple to implement and computationally effective for solving real-world optimization issues. Additionally, HSBOA's algorithmic stages are discussed as follows:

Step i): Initialization phase

Equation (10) is used to initialize the algorithm by randomly selecting the location of Secretary birds in the search space.

$$Q_{\varphi, \mu} = LB_{\mu} + L \times (UB_{\mu} - LB_{\mu}), \quad \varphi = 1, 2, \dots, v, \mu = 1, 2, \dots, q \quad (10)$$

wherein, UB_{μ} indicates upper bounds, Q_{φ} signifies area of φ^{th} secretary bird, L specifies a random number amongst 0 and 1 and LB_{μ} represents lower bound.

Equation (11) indicates that the algorithm employs a population based approach, in which a population of potential solutions is used as the initial phase for optimization.

$$Q = \begin{bmatrix} g_{1,1} & g_{1,2} & \cdots & g_{1,\mu} & \cdots & g_{1,q} \\ g_{2,1} & g_{2,2} & \cdots & g_{2,\mu} & \cdots & g_{2,q} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ g_{\varphi,1} & g_{\varphi,2} & \cdots & g_{\varphi,\mu} & \cdots & g_{\varphi,q} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ g_{v,1} & g_{v,2} & \cdots & g_{v,\mu} & \cdots & g_{v,q} \end{bmatrix}_{v \times q} \quad (11)$$

wherein, total amount of group member (secretary) is symbolized as v , g_{φ} signifies φ^{th} secretary bird, secretary bird group is represented by Q , q is a representation of the variable dimension problem, and $g_{\varphi,\mu}$ indicates φ^{th} secretary bird for the μ^{th} query the value of the variable.

Step ii): Fitness function

The objective function is employed to assess how well a solution performs in optimization issues. Also, fitness function measures the quality of solution by assigning it a fitness score, which guides the search for optimum solutions. The expression for fitness function is specified in equation (9).

Step iii): Position updation

Two distinct natural behaviours are employed to apprise HSBOA members. Both of these categories, such as secretary bird escape and hunting techniques. Therefore, during every iteration, secretary bird colony's each member obtains two distinct updates.

Step iv): Secretary bird's hunting technique

When feeding on snakes, secretary birds exhibit three distinct stages of their hunting behavior, like consuming prey, seeking prey, and attacking prey.

a) Searching for prey: Equation (12) can be exploited to create a mathematical framework of the secretary bird's position updates whereas it searches for prey.

$$\text{while } j < \frac{1}{3}H, g_{\varphi,\mu}^{newe_1} = g_{\varphi,\mu} + (g_{rand_1} - g_{rand_2}) \times \Omega_1 \quad (12)$$

In the primary phase, Ω_1 symbolizes a randomly formed array of dimension $1 \times q$ from interval $[0,1]$, $g_{\varphi,\mu}^{newe_1}$ specifies new phase of φ^{th} secretary bird at μ^{th} size, g_{rand_1} and g_{rand_2} indicates random candidate solutions, extreme iteration is specified by H , the present iteration count is symbolized by j , and $g_{\varphi}^{newe_1}$ designates the new phase of φ^{th} secretary bird.

b) Consuming prey: After catching a snake, a secretary bird develops a special hunting approach. The secretary bird frequently stops eating its prey to use its excellent vision to locate the snake. At this stage, random movements of the secretary bird are simulated using Brownian motion (RB). Brownian motion can be precisely modelled exploiting equation (13).

$$RB = randn(1, q) \quad (13)$$

During the stage of consuming prey, the secretary bird's position can be updated statistically exploiting equation (14).

$$\text{while } \frac{1}{3}H < j < \frac{2}{3}H, g_{\varphi,\mu}^{new P_1} = g_{best} + \exp((j/H)^4) \times (RB - 0.5) \times (g_{best} - g_{\varphi,\mu}) \quad (14)$$

wherein, g_{best} stands for current finest value, and $randn(1, q)$ symbolizes an array of dimension $1 \times q$ that is produced at random using a standard normal distribution.

c) Attacking prey: The secretary bird typically employs its leg kicking method, which is done by raising its leg with precision and delivering a sharp kick by targeting the snake's head with its talons. The random search procedure employs the Levy flight method to improve the optimizer's ability to search globally, decrease the possibility that the algorithm will become stuck in local solutions, and boost the algorithm's convergence and accuracy. During the attacking prey phase, the secretary bird's position can be updated scientifically utilizing equation (15).

$$\text{while } j > \frac{2}{3}H, g_{\varphi,\mu}^{new P_1} = g_{best} + \left(\left(1 - \frac{j}{H} \right) \wedge \left(2 \times \frac{j}{H} \right) \right) \times g_{\varphi,\mu} \times b \quad (15)$$

Here, b is used to represent the weighted Levy flight for increasing the optimization accuracy of the algorithm.

$$b = 0.5 \times Levy(q) \quad (16)$$

From the above equation, Levy flight distribution function is indicated by $Levy(q)$.

Step v): Secretary bird's escaping method

The evasion approaches can be generally classified into two main categories. Running swiftly or taking flight is the first strategy, and camouflage is the second. First, when secretary birds sense the presence of a predator, they begin by searching for a suitable environment to camouflage themselves. Secretary birds can use structural elements or colours to blend in with their environment, making it hard for predators to locate them. Expression (17) provides a mathematical model for both the evasion tactics employed by secretary birds.

$$g_{\varphi,\mu}^{new P_1} = \begin{cases} \alpha_1 : g_{best} + (2 \times RB - 1) \times \left(1 - \frac{j}{H} \right) \times g_{\varphi,\mu}, & \text{if } rand < M_\varphi \\ \alpha_2 : g_{\varphi,\mu} + \Omega_2 \times (g_{rand} - \gamma \times g_{\varphi,\mu}), & \text{else} \end{cases} \quad (17)$$

In the above expression, $M = 0.5$, a random group of an array of dimension $(1 \times q)$ from an ordinary distribution is denoted by Ω_2 , g_{rand} stands for the current iteration's random candidate solution, and γ represents the randomly chosen among 1 or 2, which is measured employing equation (18).

$$\gamma = \text{round}(1 + \text{rand}(1,1)) \quad (18)$$

The $\text{rand}(1,1)$ function generates a random number among 0 and 1.

Considering the second condition from equation (18) at iteration y , we get

$$g_{\varphi,\mu}(y+1) = g_{\varphi,\mu}(y) + \Omega_2 \times (g_{\text{rand}} - \gamma \times g_{\varphi,\mu}(y)) \quad (19)$$

$$g_{\varphi,\mu}(y+1) = g_{\varphi,\mu}(y) + \Omega_2 \times g_{\text{rand}} - \Omega_2 \gamma \times g_{\varphi,\mu}(y) \quad (20)$$

$$g_{\varphi,\mu}(y+1) = g_{\varphi,\mu}(y)[1 - \Omega_2 \gamma] + \Omega_2 * g_{\text{rand}} \quad (21)$$

To find unknown deterministic elements, time series analysis is combined with various test procedures. When modelling a time series, it is necessary to take into account its periodic structure. As explained below, the periodic structure is taken into account for time series modelling based on harmonic analysis [22].

$$g_1, g_2, g_3, \dots, g_y, \dots, g_i \quad (22)$$

Time series data at the y^{th} instant under consideration is specified as g_y , and series length is represented by i .

The series g_y is described by the orthogonal trigonometric function for each module. $i = 2F + 1$ if i is odd, and the equation is given below,

$$g_y = z_0 + \sum_{x=1}^F \left(z_x \cos\left(\frac{2\pi xy}{i}\right) + C_x \sin\left(\frac{2\pi xy}{i}\right) \right) \quad (23)$$

In the above equation, x indicates whole cycles, and preselected constants are symbolized as z and C .

Contemplate, $i = 2$ and $F = 1$. Then, the expression (23) can be written as follows:

$$g_y = z_0 + z_1 \cos\left(\frac{2\pi y}{2}\right) + C_1 \sin\left(\frac{2\pi y}{2}\right) \quad (24)$$

$$g_y = z_0 + z_1 \cos \pi y + C_1 \sin \pi y \quad (25)$$

where,

$$z_0 = \frac{1}{i} \sum_{y=1}^i g_y \quad (26)$$

$$z_0 = \frac{1}{2} [g_1 + g_2] \quad (27)$$

$$z_x = \frac{2}{i} \sum_{y=1}^i g_y \cos \left(\frac{2\pi xy}{i} \right) \quad (28)$$

Consider, $x = 1$, $i = 2$, then equation (28) can be articulated beneath,

$$z_1 = \frac{2}{2} \left[g_1 \cos \left(\frac{2\pi}{2} \right) + g_2 \cos \left(\frac{2\pi 2}{2} \right) \right] \quad (29)$$

$$z_1 = g_1 \cos(\pi) + g_2 \cos(2\pi) \quad (30)$$

When the values of $\cos \pi$ and $\cos 2\pi$ are applied in expression (30),

$$z_1 = g_1(-1) + g_2(1) \quad (31)$$

Also, C_x is signified by the formula as,

$$C_x = \frac{2}{i} \sum_{y=1}^i g_y \sin \left(\frac{2\pi xy}{i} \right) \quad (32)$$

Contemplate, $x = 1$, $i = 2$ then,

$$C_1 = \frac{2}{2} \left[g_1 \sin \left(\frac{2\pi}{2} \right) + g_2 \sin \left(\frac{2\pi 2}{2} \right) \right] \quad (33)$$

$$C_1 = g_1 \sin \pi + g_2 \sin 2\pi \quad (34)$$

When the values of $\sin \pi$ and $\sin 2\pi$ are substituted in equation (34),

$$C_1 = g_1(0) + g_2(0) \quad (35)$$

Assume time series is implied as $g(y-1)$, $g(y)$ and $g(y+1)$. Take $g_1 = g(y-1)$, $g_2 = g(y)$, and $g_y = g(y+1)$, and substituting g_1 , g_2 and g_y values, and the expression becomes,

$$g(y+1) = z_0 + z_1 \cos \pi y + C_1 \sin \pi y \quad (36)$$

$$z_0 = \frac{1}{2} [g(y-1) + g(y)] \quad (37)$$

$$z_1 = -g(y-1) + g(y) \quad (38)$$

$$C_1 = 0 \quad (39)$$

Applying the equations (37), (38) and (39) in expression (36), we get

$$g(y+1) = \left[\frac{1}{2} [g(y-1) + g(y)] \right] + [-g(y-1) + g(y)] \cos \pi y \quad (40)$$

$$g(y+1) = \frac{1}{2} g(y-1) + \frac{1}{2} g(y) - g(y-1) \cos(\pi y) + g(y) \cos(\pi y) \quad (41)$$

$$g(y+1) = g(y) \left[\frac{1 + 2 \cos(\pi y)}{2} \right] + g(y-1) \left[\frac{1 - 2 \cos(\pi y)}{2} \right] \quad (42)$$

$$g(y) \left(\frac{1 + 2 \cos(\pi y)}{2} \right) = g(y+1) - g(y-1) \left(\frac{1 - 2 \cos(\pi y)}{2} \right) \quad (43)$$

$$g(y) = \frac{1}{1 + 2 \cos(\pi y)} \left[2g(y+1) - g(y-1)(1 - 2 \cos(\pi y)) \right] \quad (44)$$

Contemplate, $g(y)$ as $g_{\varphi, \mu}(y)$, and the expression can be written as,

$$g_{\varphi, \mu}(y) = \frac{1}{1 + 2 \cos(\pi y)} \left[2g(y+1) - g(y-1)(1 - 2 \cos(\pi y)) \right] \quad (45)$$

Substituting $g_{\varphi, \mu}(y)$ value in expression (21), and the expression can be rewritten as,

$$g_{\varphi, \mu}(y+1) = \left(\frac{2g(y+1) - g(y-1)(1 - 2 \cos(\pi y))}{1 + 2 \cos(\pi y)} \right) [1 - \Omega_2 \gamma] + \Omega_2 * g_{rand} \quad (46)$$

$$\frac{g_{\varphi, \mu}(y+1)(1 + 2 \cos(\pi y) - 2g_{\varphi, \mu}(y))}{1 + 2 \cos(\pi y)} = \frac{(\Omega_2 * g_{rand})(1 + 2 \cos(\pi y) - (g_{\varphi, \mu}(y-1)(1 - 2 \cos(\pi y))))(1 - \Omega_2 \gamma)}{1 + 2 \cos(\pi y)} \quad (47)$$

$$g_{\varphi, \mu}(y+1)[1 + 2 \cos(\pi y) - 2] = (\Omega_2 * g_{rand})(1 + 2 \cos(\pi y) - (g_{\varphi, \mu}(y-1)(1 - 2 \cos(\pi y))))(1 - \Omega_2 \gamma) \quad (48)$$

$$g_{\varphi, \mu}(y+1) = \frac{1}{2 \cos(\pi y) - 1} \left[(\Omega_2 * g_{rand})(1 + 2 \cos(\pi y) - (g_{\varphi, \mu}(y-1)(1 - 2 \cos(\pi y))))(1 - \Omega_2 \gamma) \right] \quad (49)$$

The updated equation for HSBOA is specified in equation (49).

Step vi): Fitness evaluation

Once the secretary bird's position has been changed, Equation (9) is used to calculate each member's fitness, and the secretary bird with minimal fitness is chosen as the finest choice.

Step vii): Termination

The aforementioned procedures are repeated until the maximal iteration value is attained. In Algorithm 1, the HSBOA pseudocode is displayed.

Algorithm 1. Pseudocode of HSBOA

Initialize population
For $j=1:H$
Appraise secretary bird g_{best}
For $\varphi=1:v$
Exploration phase:
If $j < \frac{1}{3}H$
Compute novel position of φ^{th} secretary bird employing eqn (12)
else if $\frac{1}{3}H < j < \frac{2}{3}H$
Estimate innovative location of φ^{th} secretary bird through eqn (14)
else
Analyse novel position of φ^{th} secretary bird employing eqn (15)
end if
Exploitation stage:
If $rand < 0.5$
Estimate novel location of φ^{th} secretary bird α_1 by equation (17)
else
Calculate novel position of φ^{th} secretary bird α_2 in equation (17)
end if
end for $\varphi=1:v$
Save finest candidate solution
end for $j=1:H$
Outcome: Optimal solution gained utilizing HSBOA
Return optimal solution

HSBOA improves the ability to escape local minima and better explore the solution space, improving the entire technique's accuracy and performance. The training procedure becomes more effective, leading to faster convergence by employing HSBOA, leading to better feature selection ability. The selected features ζ_{fs} are then subjected to the QRNN for intrusion detection.

4.4 Intrusion detection using proposed Harmonic Secretary Bird Optimization Algorithm_Quasi-Recurrent Neural Network

Monitoring network or system activity for malicious activity or security breaches is known as intrusion detection. It involves detecting any anomalies, abuse, or illegal access that might indicate probable attacks or vulnerabilities in the system. Designing IDS that can efficiently distinguish between safe user behavior and possible threats in real-time, reduce false positives, adjust to new and developing attack methods, and scale to manage massive data volumes with minimal computational overhead is still a challenge. Developing an IDS that can offer efficient, precise, and effective defense against increasingly complex cyber threats requires addressing these problems. The proposed HSBOA_QRNN is used in this paper to detect intrusions. Harmonic analysis [22] and SBOA [20] are combined to create the proposed HSBOA algorithm, which is used here to train the QRNN [19] used for detecting intrusions. Here, the selected feature ζ_{fs} attained from the SVM-RFE are subjected as input to the QRNN. Below is an explanation of QRNN's architecture and training process:

4.4.1 Architecture of QRNN

Here, intrusion detection is done employing the QRNN architecture. QRNN is a hybrid neural network encouraged by LSTM, combining the advantages of both CNNs and LSTMs [19]. Long-range dependency can be modelled by LSTM, while CNN is rapid and highly parallelizable. The convolutional layers in QRNNs speed up training times by reducing the need for sequential computations, which increases parallelization. QRNNs are adaptable and can be used for a variety of sequence-based tasks by varying the depth and size of recurrent components and convolutional layers. The hidden layer's outcome at the time $\psi(N_\psi)$ is actually causing an execution bottleneck because it depends on $N_{\psi-1}$, which subsequently depends on $N_{\psi-2}$ etc. For each recurrent step ψ , QRNN reduces the computational load by removing the LSTM's hidden-hidden matrix multiplication and providing a convolution dimension of $\tilde{h}$ over input h , which is the selected feature ζ_{fs} . Additionally, the forget gate E_ψ and outcome gate $\mathcal{G}_\psi$ are produced by concerning input and forget gates. The subsequent is an equation for the procedure that was developed in the QRNN with f_0 -pooling:

$$E_\psi, \mathcal{G}_\psi = \delta \cdot (\kappa_{E,g} \cdot h_{\psi-\tilde{h}+1;\psi}) \quad (50)$$

$$\tilde{t}_\psi = \tanh(\kappa_t \cdot h_{\psi-\tilde{h}+1;\psi}) \quad (51)$$

$$t_\psi = t_{\psi-1} \odot E_\psi + \tilde{t}_\psi \odot (1 - E_\psi) \quad (52)$$

$$N_\psi = t_\psi \odot \mathcal{G}_\psi \quad (53)$$

wherein, t denotes candidate state, weight is signified by κ and activation function indicated as δ . The recurrent step can be implemented more quickly by using a gated or weighted sum. This ensures a likelihood for an unbounded activation function, such as ReLU function, by preventing hidden layer from growing rapidly. The acquired output from the QRNN technique is designated by ω_{out} . The graphical representation of QRNN framework is illustrated in Figure 3.

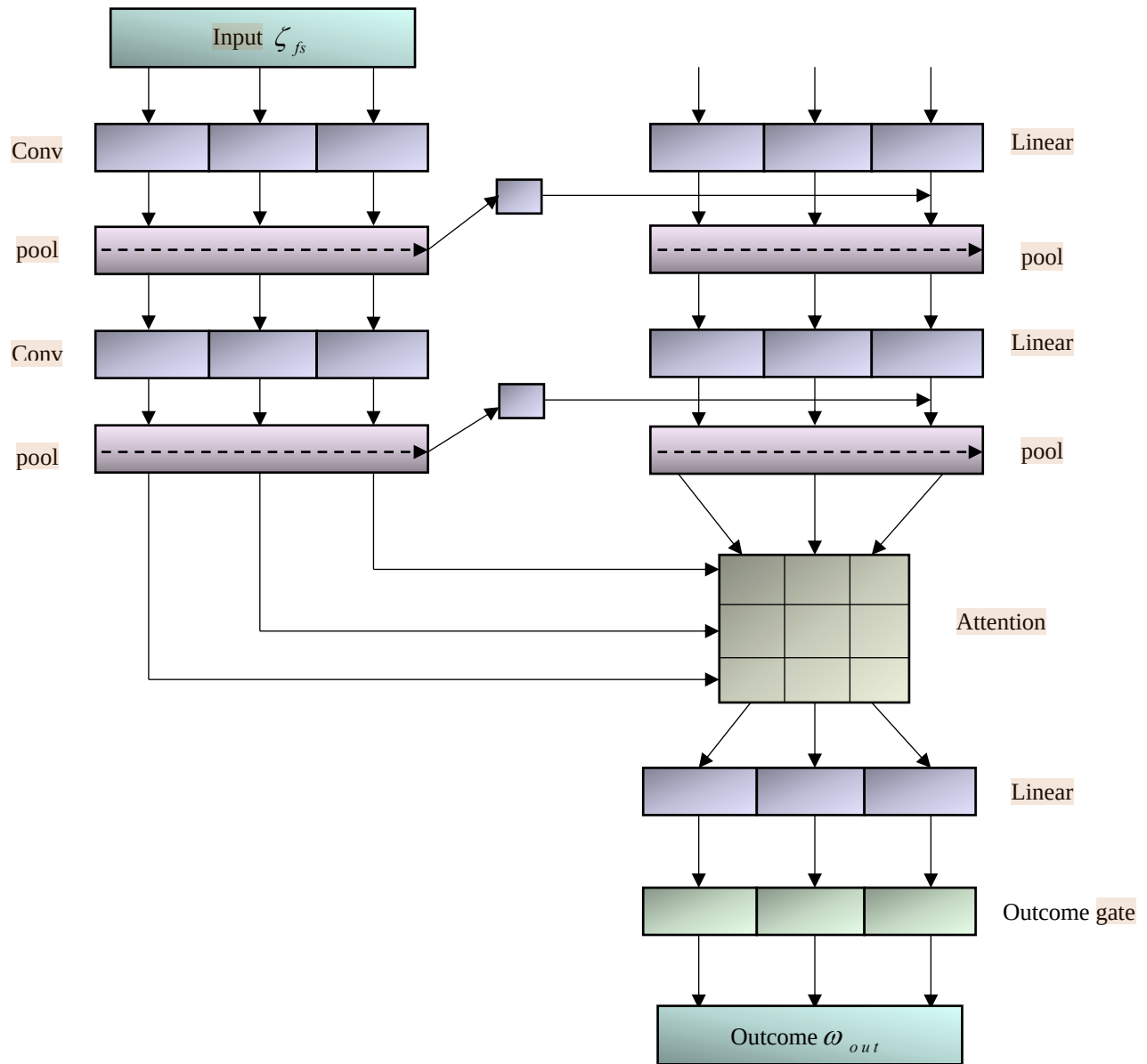


Figure 3. Systematic representation of the QRNN

4.4.2 Devised Harmonic Secretary Bird Optimization Algorithm for training of Quasi-Recurrent Neural Network

The performance of the model is optimized by efficiently adjusting the parameters using the HSBOA algorithm to update the QRNN's weights, leading to improved accuracy and faster convergence. Here, the Harmonic analysis [22] and the SBOA [20] are combined to create the HSBOA algorithm. Although section 4.3.1 provides a detailed explanation of the presented HSBOA algorithm, the fitness parameter is different and is written as,

$$MSE_{Fitness} = \frac{1}{S} \sum_{l=1}^S [\omega_{outl} - \omega_{outl}^*]^2 \quad (54)$$

Here, the number of data points is signified by S , the observed outcome from QRNN is specified as ω_{out} and predicted value is symbolized as ω_{out}^* .

5. Results and discussion

A detailed examination of the outcomes acquired during the implementation of the designed IDS is provided in this section, along with dataset used, experimental settings, and comparative discussions.

5.1 Experimental setup

The performance of the designed HSBOA-QRNN is implemented in the Python tool employing the ISCX NSL-KDD dataset 2009 [21].

5.2 Dataset description

The NSL-KDD dataset [21] is a widely used dataset for evaluating network intrusion detection systems. It consists of multiple files in different formats and subsets. The KDDTrain+.ARFF file contains the full NSL-KDD training set with binary labels in ARFF format, while the KDDTrain+.TXT file provides the same training data in CSV format with attack-type labels and difficulty levels. A 20% subset of the KDDTrain+ dataset is available in both ARFF (KDDTrain+_20Percent.ARFF) and TXT (KDDTrain+_20Percent.TXT) formats. For testing, the KDDTest+.ARFF and KDDTest+.TXT files contain the complete test set, with the ARFF version having binary labels and the TXT version providing attack-type labels and difficulty levels. Additionally, the KDDTest-21. ARFF and KDDTest-21.TXT files are subsets of the KDDTest+ set, excluding records with a difficulty level of 21, offering a refined test set for different evaluation scenarios.

5.3 Evaluation measures

Evaluation parameters, namely TNR, accuracy, and TPR are taken into consideration when evaluating the efficiency of the designed HSBOA-QRNN.

a) TNR: The percentage of real normal activities (non-intrusions) that the framework accurately detects is computed by TNR. The formula employed for calculating the TNR is given below,

$$TNR = \frac{\tau^{tn}}{\tau^{tn} + \tau^{fp}} \quad (55)$$

Here, τ^{tn} and τ^{fp} implies true negative and false positive, respectively.

b) TPR: The percentage of actual intrusions that the technique precisely detects is computed by TPR. The formula utilized for computing the TPR is given below,

$$TPR = \frac{\tau^{tp}}{\tau^{tp} + \tau^{fn}} \quad (56)$$

wherein, τ^{tp} and τ^{fn} designates true positive and false negative, correspondingly.

c) Accuracy: Accuracy assesses the model's correctness by computing the proportion of precise predictions to overall predictions. The computation expression for accuracy is designated beneath,

$$Accuracy = \frac{\tau^{tp} + \tau^{tn}}{\tau^{tp} + \tau^{tn} + \tau^{fp} + \tau^{fn}} \quad (57)$$

d) Training loss: The training loss is computed exploiting the 'categorical_cross entropy' loss function, which is commonly utilized for multi-class classification issues.

5.4 Data visualization plot

Figure 4 displays the evaluation considering various features contained in the network traffic data from the ISCX NSL-KDD dataset 2009 employed by HSBOA_QRNN in intrusion detection. Here, the graph is depicted among feature values and samples. The value of the dst_host_same_src_port_rate ranges from 0.2 to 0.9, while the values of the srv_error_rate feature alter from 0.1 to 0.9. In addition, land values are within the range of 0.2 to 0.4, logged_in values ranging from 0.4 to 0.8, and the values of dst_host_error_rate within the range of 0 to 1, when considering several samples as 20,000.

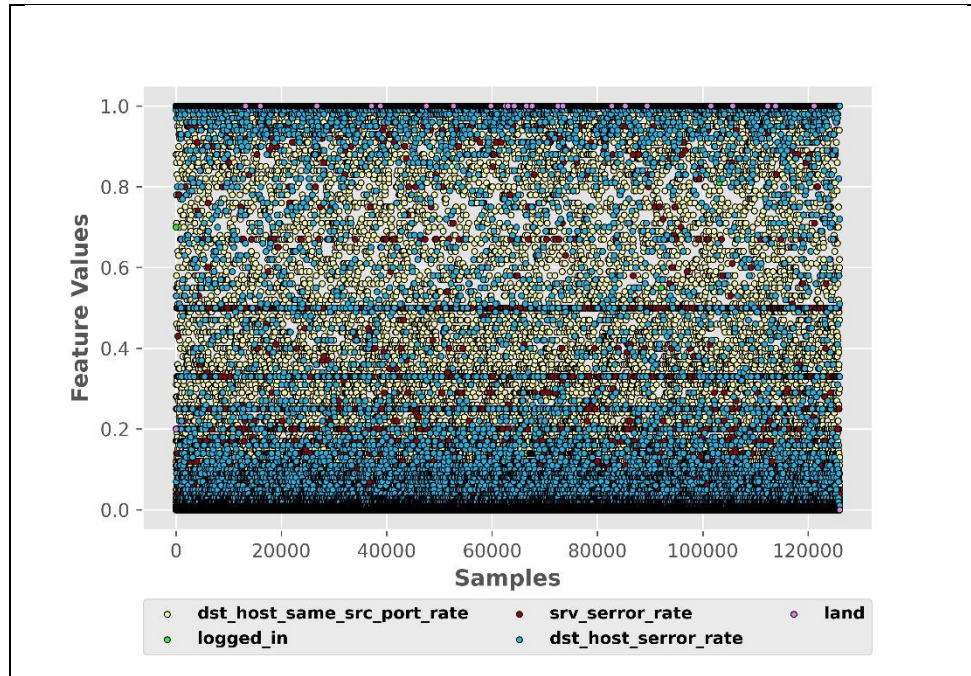


Figure 4. Data visualization plot

5.5 Training loss

Figure 5 portrays the graph plotted amongst training loss and epochs. Here, when the epoch is contemplated as 80, the value attained as training loss by the HSBOA_QRNN is 0.046454. With epochs as 60, the training loss attained is 0.064778. A gradual decrease in training loss is witnessed with the increase in amount of epochs.

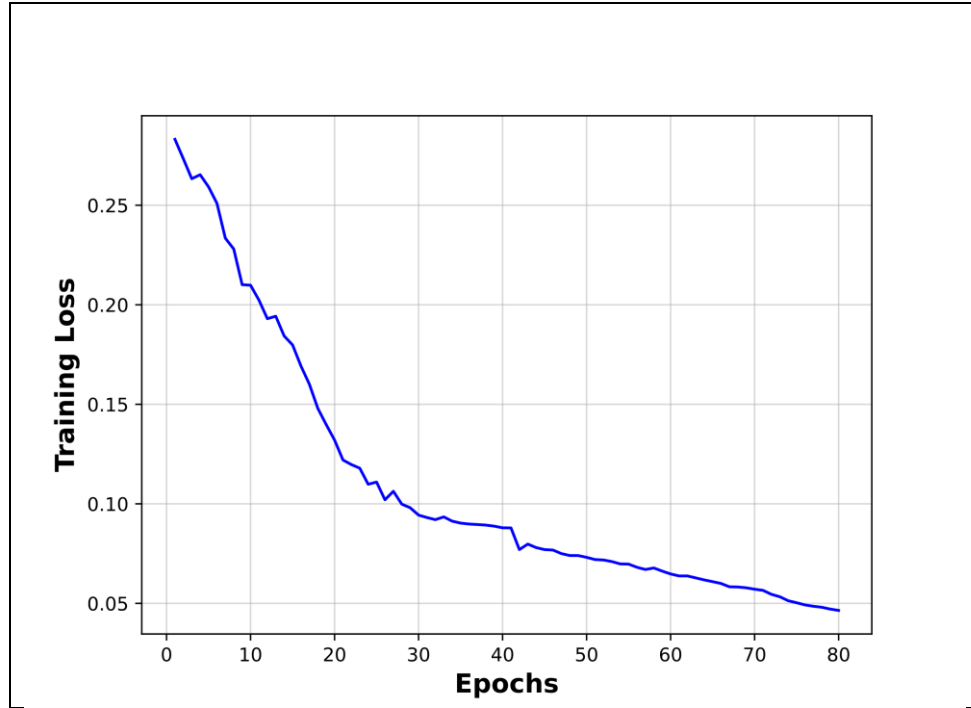


Figure 5. Training loss

5.6 Performance analysis

The efficiency of the HSBOA_QRNN devised in the present research is evaluated by changing the epoch and feature_sizes with multiple training sample percentages.

51 5.6.1 Based on varying epoch

Figure 6 portrays the analysis of the HSBOA_QRNN is evaluated with respect to evaluation parameters. Here, Figure 6a) demonstrates the accuracy-based evaluation of HSBOA_QRNN, where for the training samples of 80%, the HSBOA_QRNN approach acquired accuracy of 85.658%, 87.866%, 89.173%, and 90.185%, for the epochs as 20, 40, 60, and 80. The appraisal of HSBOA_QRNN regarding TNR is displayed in Figure 6b). The HSBOA_QRNN technique quantified TNR of 87.000% for 20 epochs, 88.602% for 40 epochs, 90.614% for 60 epochs, and 91.223% for 80 epochs for training samples of 90%. The TPR obtained by HSBOA_QRNN with varying training samples and epochs is displayed in Figure 6c), here for the epochs of 20, 40, 60, and 80, the HSBOA_QRNN gained TPR of 83.868%, 85.568%, 87.058%, and 88.569%, when contemplating the training samples of 70%.

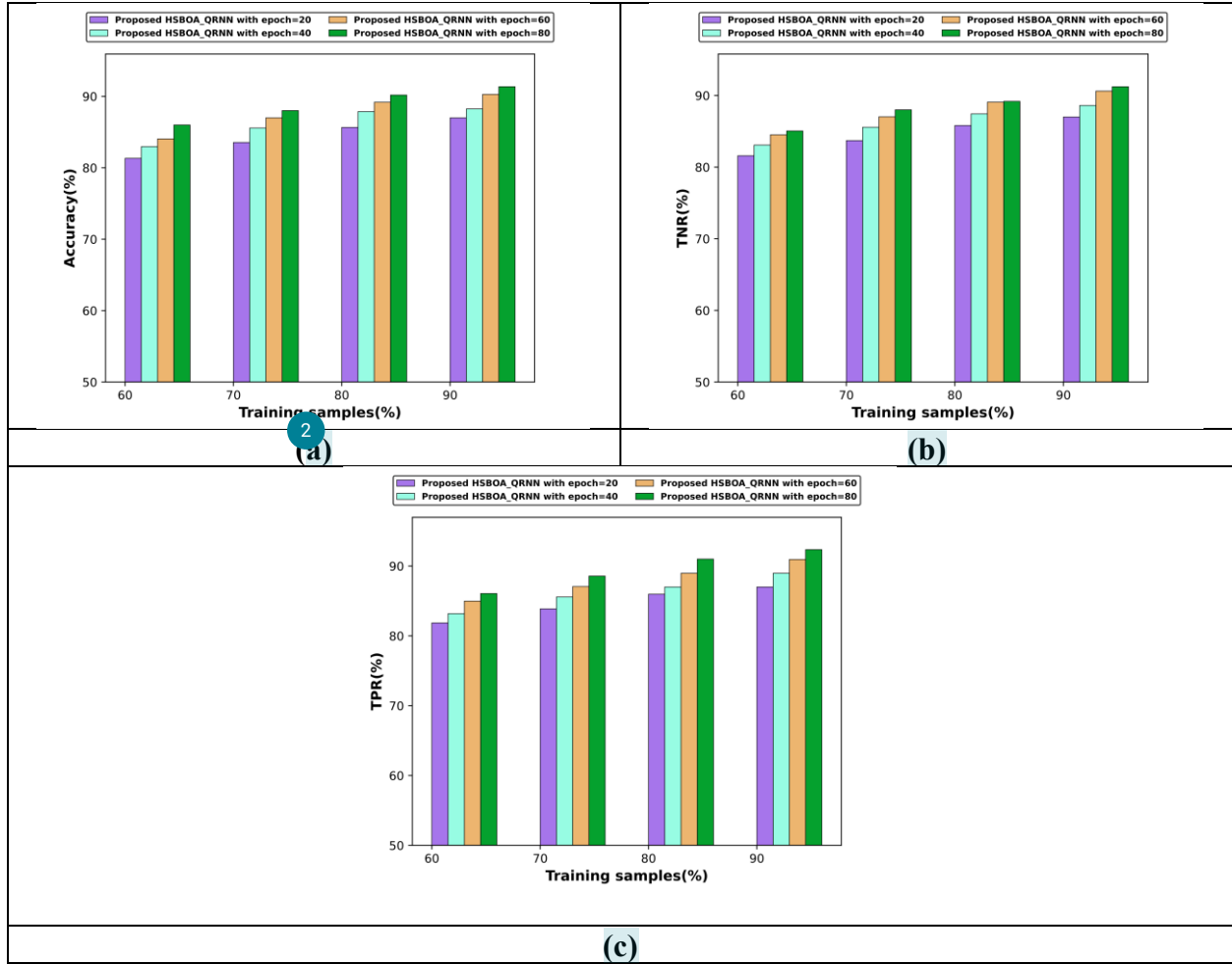


Figure 6. Investigation of the HSBOA_QRNN on the basis of a) accuracy, b) TNR, and c) TPR

5.6.2 Based on the feature_sizes

Figure 7 evaluates the effectiveness of HSBOA_QRNN using numerous feature_sizes and evaluation parameters, namely accuracy, TPR, and TNR. The evaluation of HSBOA_QRNN based upon accuracy is portrayed in Figure 7a). For 80% of training samples, the HSBOA_QRNN recorded accuracies of 85.660%, 87.646%, 89.006%, and 90.185% for feature_sizes of 4, 6, 8, and 10, respectively. The efficacy of the HSBOA_QRNN was assessed based upon TNR, and is signified in Figure 7b). At 90% training samples, the HSBOA_QRNN recorded TNR values of 87.084%, 88.602%, 90.637%, and 92.353% for feature_sizes of 4, 6, 8, and 10. The valuation of the HSBOA_QRNN regarding TPR is represented in Figure 7c). For a 60% training sample, the HSBOA_QRNN attained TPR values of 82.178%, 83.061%, 84.597%, and 86.061% for feature_sizes of 4, 6, 8, and 10.

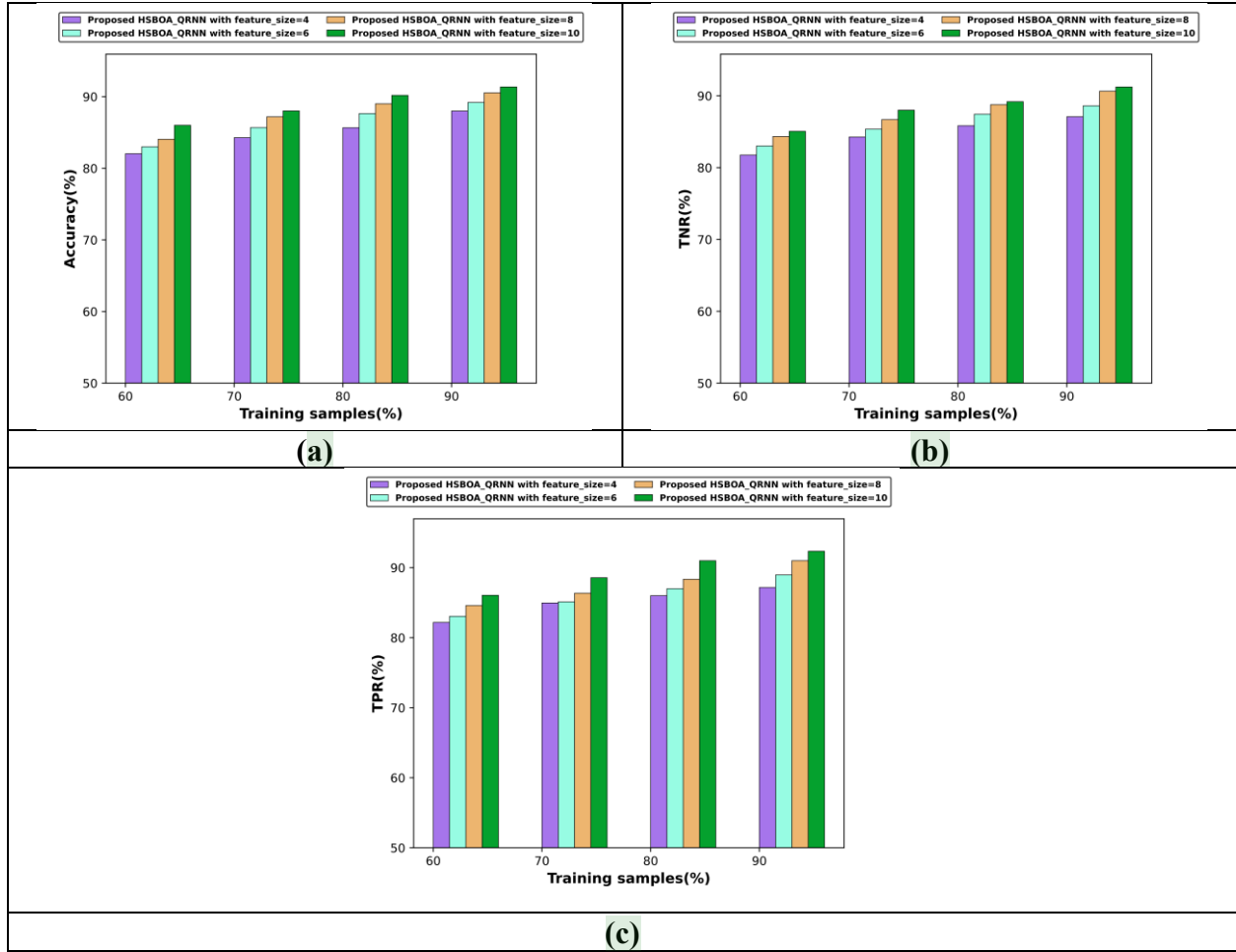


Figure 7. Valuation of the HSBOA_QRNN according to, a) accuracy, b) TNR, and c) TPR

5.7 Algorithmic techniques

To investigate the effectiveness of the HSBOA+QRNN, different algorithms, including the Crow Search Algorithm+QRNN (CSA+QRNN) [24][19], Pelican Optimization Algorithm+QRNN (POA+QRNN) [25][19], and Secretary Bird Optimization Algorithm+QRNN (SBOA+QRNN) [20][19] are considered. Further, the algorithmic examination of HSBOA+QRNN is performed by changing swarm sizes between 10 and 50.

5.8 Algorithmic analysis

Figure 8 describes the algorithmic evaluation of HSBOA+QRNN regarding various performance parameter by changing swarm size. The accuracy-related valuation of HSBOA_QRNN is demonstrated in Figure 8a). By considering swarm size as 50, the accuracy recorded by the methods, such as CSA+QRNN, POA+QRNN, SBOA+QRNN, and HSBOA+QRNN is 86.796%, 87.787%, 89.275%, and 91.358%, correspondingly. When comparing HSBOA+QRNN to CSA+QRNN, POA+QRNN, and SBOA+QRNN, the performance enhancement of HSBOA_QRNN is 4.562%, 3.571%, and 2.083%, respectively. The TNR-based analysis of HSBOA+QRNN is presented in Figure 8b). TNR measured by HSBOA+QRNN is 87.790% when swarm size is considered to be 30, whereas TNR quantified by standard approaches, like

CSA+QRNN is 82.132%, POA+QRNN is 83.677%, and SBOA+QRNN is 85.570%. In comparison to CSA+QRNN, POA+QRNN, and SBOA+QRNN, the performance enhancement of HSBOA+QRNN based on TNR is 5.658%, 4.113%, and 2.22%. The assessment of HSBOA+QRNN with regard to TPR is signified in Figure 8c). At swarm size is 40, the TPR value determined by HSBOA+QRNN is 90.999%, while the TPR value estimated by CSA+QRNN, POA+QRNN, and SBOA+QRNN is 85.576%, 86.979%, and 88.998%, respectively. Comparing HSBOA+QRNN in terms of TPR to CSA+QRNN, POA+QRNN, and SBOA+QRNN, the performance enhancement is by 5.423%, 4.02%, and 2.001%.

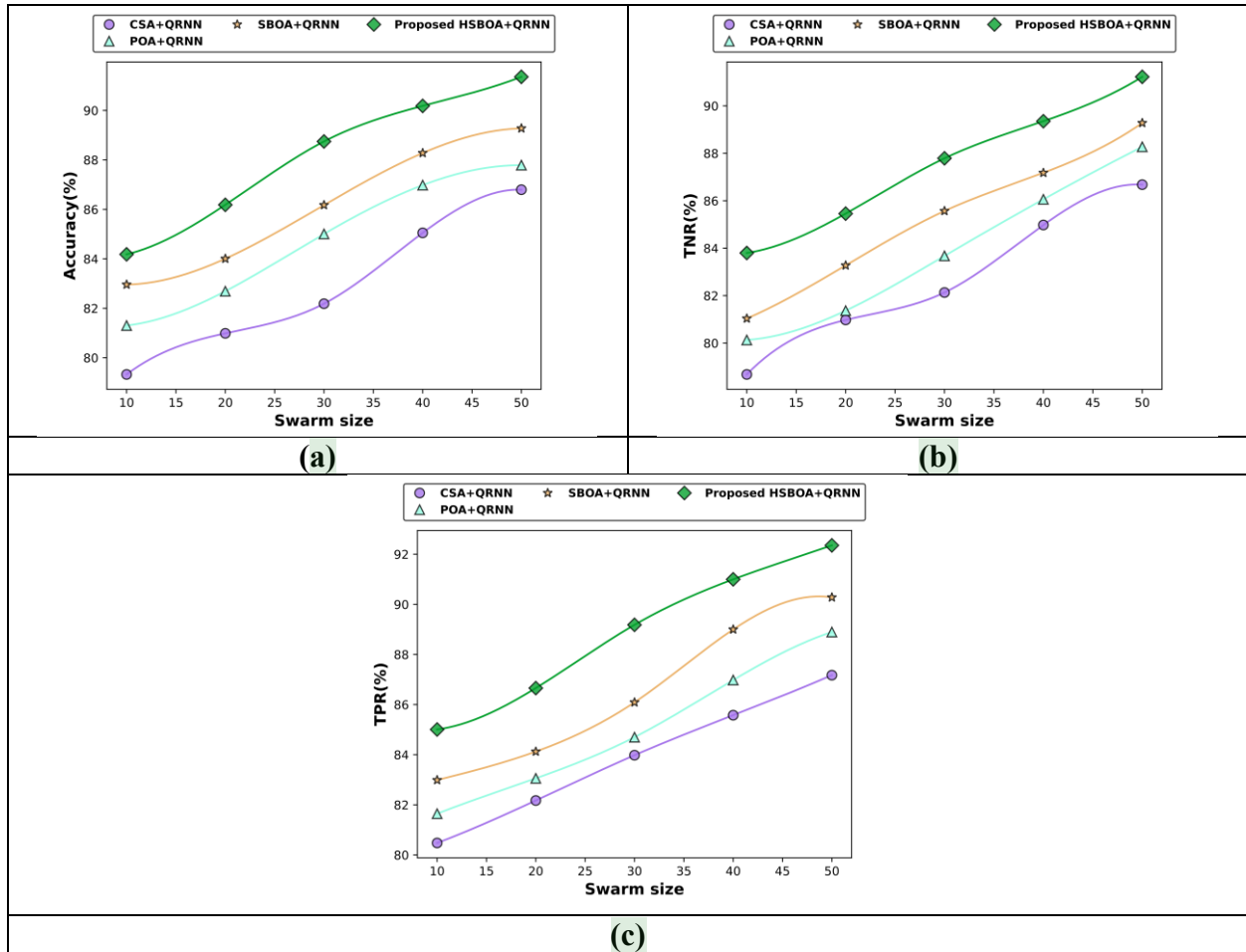


Figure 8. Evaluation of HSBOA+QRNN by changing swarm size in terms of a) accuracy, b) TNR, and c) TPR

5.9 Comparative methods

The performance of the HSBOA_QRNN for cloud computing intrusion detection described in this paper is evaluated by contrasting it with techniques like CybS-CC-SACGAN-COA [1], RF-RBFNN [3], FEFS-DLM [5], and IMFL-IDSCS [6].

5.10 Comparative analysis

Figure 9 illustrates the evaluation of HSBOA_QRNN, considering various measures while varying training samples from 60% to 90%. Specifically, Figure 9a) displays the accuracy analysis of the

HSBOA_QRNN. The accuracy computed by HSBOA_QRNN is 90.185%, while baseline models, namely CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM and IMFL-IDSCS acquired accuracy of 82.467%, 84.057%, 85.980%, and 88.178%, for training sample as 80%. HSBOA_QRNN has attained maximum accuracy, which is 6.128% higher than RF-RBFNN. The investigation of HSBOA_QRNN concerning TNR is specified in Figure 9b). By contemplating the training sample of 60%, the TNR acquired by HSBOA_QRNN is 85.051%, whereas prevailing models, such as CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM, and IMFL-IDSCS quantified TNR of 77.877%, 79.686%, 81.965%, and 83.273%. The HSBOA_QRNN model has acquired a higher TNR superior by 3.086% than the FEFS-DLM approach. Figure 9c) portrays the analysis of HSBOA_QRNN regarding TPR. In addition, the traditional models, including CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM, and IMFL-IDSCS attained TPR of 84.005%, 85.274%, 86.599%, and 88.573%, by 80% training sample, whereas the TPR achieved by HSBOA_QRNN is 90.998%. HSBOA_QRNN has attained superior accuracy, which is 4.399% higher than FEFS-DLM.

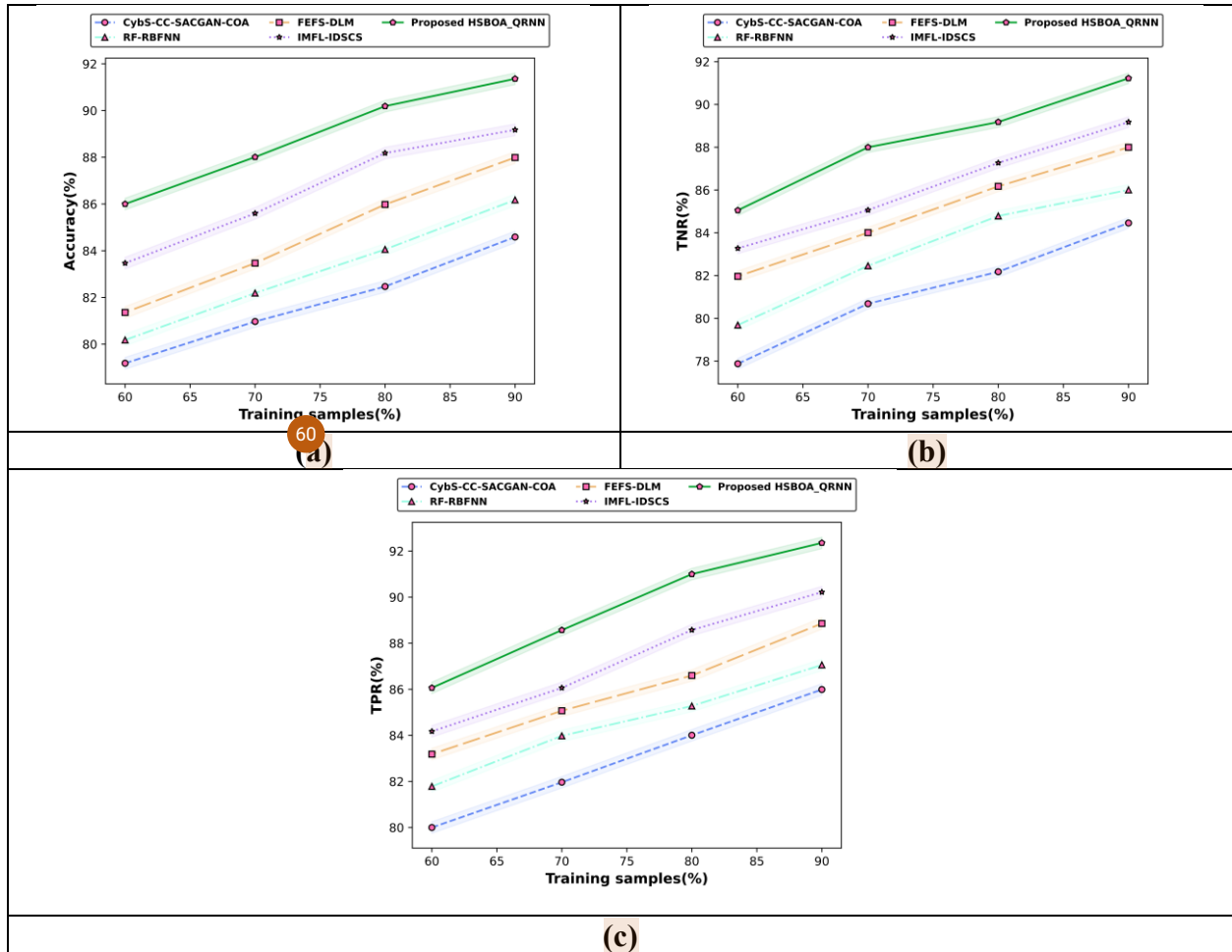


Figure 9. Evaluation of HSBOA_QRNN concerning training sample based upon, a) accuracy, b) TNR, and c) TPR

5.11 Comparative discussion

Table 1 demonstrates the comparison of the outcomes of HSBOA_QRNN with existing methods. Furthermore, HSBOA_QRNN achieved the superior accuracy, TNR, and TPR of 91.358%, 92.353%, and 91.223%, respectively, when considering 90% of training samples. Nevertheless, prevailing frameworks, namely CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM, and IMFL-IDSCS calculated accuracy of 84.588%, 86.174%, 87.988%, and 89.174%. Similarly, baseline models, like CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM, and IMFL-IDSCS acquired TNR of 85.990%, 87.056%, 88.856%, and 90.217%. Further, traditional approaches, including CybS-CC-SACGAN-COA, RF-RBFNN, FEFS-DLM, and IMFL-IDSCS gained TPR of 84.457%, 86.007%, 87.996%, and 89.174%. Consequently, HSBOA_QRNN has recorded a higher accuracy and is suitable for environments with limited resources because it uses fewer parameters. Furthermore, by incorporating harmonic analysis into SBOA, HSBOA effectively increased the convergence rate and produced best weights and bias for QRNN in a minimal amount of time, increasing QRNN's effectiveness in detecting intrusions in cloud computing.

Table 1. Comparative discussion of HSBOA_QRNN

Assessment based on	Metrics/Methods	CybS-CC-SACGAN-COA	RF-RBFNN	FEFS-DLM	IMFL-IDSCS	Proposed HSBOA_QRNN
Training sample is 90%	Accuracy (%)	84.588	86.174	87.988	89.174	91.358
	TNR (%)	85.990	87.056	88.856	90.217	92.353
	TPR (%)	84.457	86.007	87.996	89.174	91.223

6. Conclusion

The increasing complexity and scale of cloud computing environments make them more vulnerable to various cyber threats. Conventional IDS often find it difficult to adjust to the distributed and dynamic nature of cloud infrastructures, necessitating the development of more precise, scalable, and efficient models to recognize and minimize changing security threats instantly. This manuscript develops a novel optimization-based DL approach termed HSBOA_QRNN for intrusion detection. Originally, cloud simulation is conducted. Thereafter, input network traffic data is attained from the given dataset and is subjected to data preprocessing. The data preprocessing is performed by utilizing missing value restoration and data normalization employing min-max normalization. Subsequently, feature selection is carried out by exploiting SVM-RFE, where SVM is optimally trained by employing the proposed HSBOA. The HSBOA is designed by assimilating harmonic analysis and SBOA. Lastly, the intrusion is detected exploiting a QRNN, which is also trained utilizing HSBOA. The detected outcome is categorized as DoS, Probe, R2L, U2R, and Normal. Furthermore, the HSBOA_QRNN obtained a superior accuracy of 91.358%, TNR of 92.353%, and TPR of 91.223%. In the future, hybrid DL networks will be fused to boost the efficiency of intrusion detection.

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